컴퓨터와화학

제4장 분자동역학 모의실험의 원리

중요 학습 내용

- **Newton** 운동방정식
- 힘장 **(force fields)**
- **Newton** 운동방정식의 적분
- 분자동역학 모의실험의 초기조건 설정
- **Periodic boundary conditions and minimum image conventions**

계산화학의 중요성

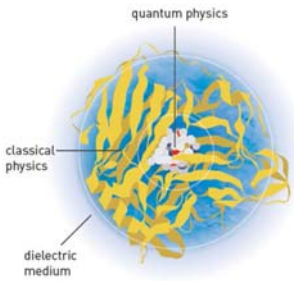
The Nobel Prize in Chemistry 2013


Photo: A. Mahmoud
Martin Karplus
Prize share: 1/3


Photo: A. Mahmoud
Michael Levitt
Prize share: 1/3


Photo: A. Mahmoud
Arieh Warshel
Prize share: 1/3

The Nobel Prize in Chemistry 2013 was awarded jointly to Martin Karplus, Michael Levitt and Arieh Warshel *"for the development of multiscale models for complex chemical systems"*.



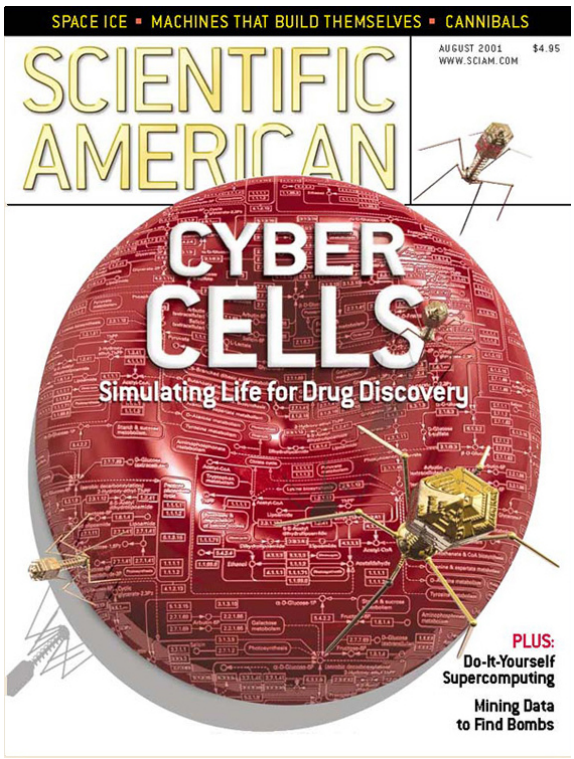
ACS National Meeting 2016 focuses on Computer & Chemistry



재료과학과 계산화학



생명과학과 계산화학



"Computational Chemistry confirms that new antimicrobial kills *E.Coli* within 30 seconds", *Science Daily, 2016*



전자구조계산과 분자동역학 모의실험

- 전자구조계산

$$\hat{H}\Psi = E\Psi$$

- 분자동역학 모의실험

$$F = ma$$

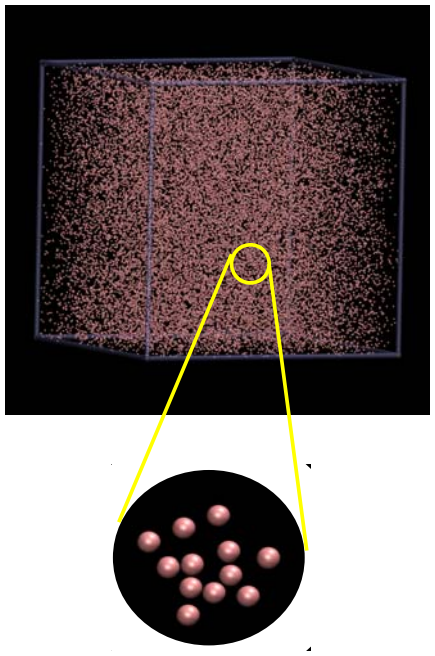
Newton 운동방정식

- Total energy of the system

$$E = \sum_{i=1}^{N} \frac{1}{2} m_i v_i^2 + V(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N)$$

- Newtonian equation of motion

$$m_i \frac{d^2 \mathbf{r}_i}{dt^2} = -\nabla V(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N) = F(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N)$$



$$V(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N) = V^{\text{bonded}}(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N) + V^{\text{non-bonded}}(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N)$$

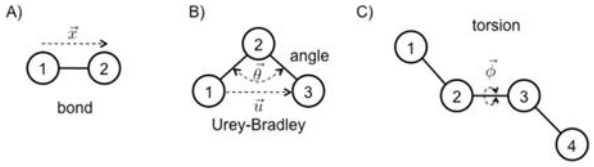
결합 상호작용에 대한 퍼텐셜 에너지

- Potential energy for bonded interaction

$$V^{\text{bonded}}(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N) = V^{\text{stretching}} + V^{\text{bending}} + V^{\text{torsion}} + V^{\text{improper}} + \cdots$$

- Stretching interaction energy

$$V^{\text{stretch}} = \sum_i k_{i,\text{stretch}}(r_i - r_{i,0})^2$$



- Bending interaction energy

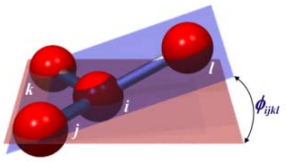
$$V^{\text{bend}} = \sum_i k_{i,\text{bend}}(\theta_i - \theta_{i,0})^2 + \sum_j k_{\text{UB}}(u_i - u_{i,0})^2$$

- Torsional interaction energy

$$V^{\text{torsion}} = \sum_i V_{i,\text{torsion}}\left[1 + \cos(n\phi_i - \phi_{i,0})\right]$$

- Improper interaction

$$V^{\text{improper}} = \sum_i k_{i,\text{improper}}(\phi_{ijkl} - \phi_{ijkl,0})^2$$



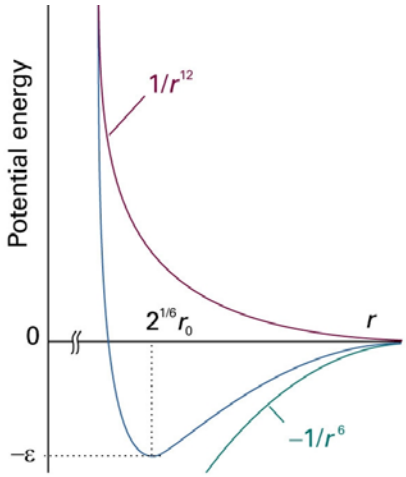
Potential energy for non-bonded interactions

- Potential energy for non-bonded interaction

$$V^{\text{non-bonded}}(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N) = V^{\text{LJ}} + V^{\text{elect}}$$

- Lennard-Jones potential energy for a monoatomi

$$V^{LJ}(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N) = \sum_{i=1}^{N}\sum_{j>i}^{N} 4\epsilon\left[\left(\frac{\sigma}{r_{ij}}\right)^{12} - \left(\frac{\sigma}{r_{ij}}\right)^{6}\right]$$



- Electrostatic potential energy

$$V^{elect}(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N) = \sum_{i=1}^{N}\sum_{j>i}^{N} \frac{1}{4\pi\epsilon_0}\frac{q_i q_j}{r_{ij}}$$

where

$$\frac{1}{r_{ij}} = \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}$$

	ϵ (kcal/mol)	σ (Å)
He	0.02027	2.28
Ne	0.09340	2.72
Ar	0.2381	3.41
Kr	0.3259	3.83

힘장 (force fields)의 종류

1. CHARMM force fields
2. AMBER force fields
3. GROMOS force fields
4. OPLS force fields
5. MM force fields
6. And so on.....

Newton 운동방정식의 적분

- Methods for numerical integrations of Newtonian equations of motion

1) Verlet algorithm

$$\mathbf{r}_i(t+\delta t) \approx 2\mathbf{r}_i(t) - \mathbf{r}_i(t-\delta t) + \frac{\mathbf{F}_i(t)}{m_i}\delta t^2 + O(\delta t^4)$$

$$\mathbf{v}_i(t) \approx \frac{\mathbf{r}_i(t+\delta t) - \mathbf{r}_i(t-\delta t)}{2\delta t} + O(\delta t^2)$$

2) Leapfrog form of Verlet algorithm

$$\mathbf{r}_i(t+\delta t) \approx \mathbf{r}_i(t) + \mathbf{r}_i(t+\frac{\delta t}{2})\delta t + O(\delta t^4)$$

$$\mathbf{v}_i(t+\frac{\delta t}{2}) \approx \mathbf{v}_i(t-\frac{\delta t}{2}) + \frac{\mathbf{F}_i(t)}{m_i}\delta t + O(\delta t^3)$$

3) Velocity Verlet algorithm

$$\mathbf{r}_i(t+\delta t) \approx 2\mathbf{r}_i(t) + \mathbf{v}_i(t)\delta t + \frac{1}{2}\frac{\mathbf{F}_i(t)}{m_i}\delta t^2 + O(\delta t^3)$$

$$\mathbf{v}_i(t+\delta t) \approx \mathbf{v}_i(t) + \frac{1}{2m_i}\big(\mathbf{F}_i(t) + \mathbf{F}_i(t+\delta t)\big)\delta t + O(\delta t^3)$$

4) Gear's predictor-corrector algorithm

Newton 운동방정식의 적분
-Velocity Verlet algorithm (1)

- Taylor expansion

$$f(x+h) = f(x) + f'(x)h + \frac{f''(x)}{2!}h^2 + \frac{f^{(3)}(x)}{3!}h^3 + \cdots$$

- Advancing the position of a particle

$$\begin{aligned}\mathbf{r}_i(t+\delta t) &= \mathbf{r}_i(t) + \mathbf{r}_i'(t)\delta t + \frac{1}{2}\mathbf{r}_i''(t)\delta t^2 + \cdots \\ &\approx \mathbf{r}_i(t) + \mathbf{v}_i(t)\delta t + \frac{1}{2}\mathbf{a}_i(t)\delta t^2 \\ &\approx \mathbf{r}_i(t) + \mathbf{v}_i(t)\delta t + \frac{1}{2}\frac{\mathbf{F}_i(t)}{m_i}\delta t^2\end{aligned}$$

$$\boxed{\mathbf{r}_i(t+\delta t) = \mathbf{r}_i(t) + \mathbf{v}_i(t)\delta t + \frac{1}{2}\frac{\mathbf{F}_i(t)}{m_i}\delta t^2 + O(\delta t^3)}$$

Newton 운동방정식의 적분
-Velocity Verlet algorithm (2)

- Advancing the velocity of a particle

$$\mathbf{v}_i(t+\frac{1}{2}\delta t) = \mathbf{v}_i(t) + \frac{1}{2}\mathbf{v}_i'(t)\delta t + \frac{1}{4}\mathbf{v}_i''(t)\delta t^2 + \cdots \approx \mathbf{v}_i(t) + \frac{1}{2}\mathbf{a}(t)\delta t + \frac{1}{4}\mathbf{v}_i''(t)\delta t^2$$
$$\approx \mathbf{v}_i(t) + \frac{1}{2}\frac{\mathbf{F}_i(t)}{m_i}\delta t + \frac{1}{4}\mathbf{v}_i''(t)\delta t^2$$

$$\begin{aligned}\mathbf{v}_i(t+\frac{1}{2}\delta t) &= \mathbf{v}_i(t+\delta t) - \frac{1}{2}t\mathbf{v}_i'(t+\delta t)\delta + \frac{1}{4}\mathbf{v}_i''(t+\delta t)\delta t^2 + \cdots \\ &\approx \mathbf{v}_i(t+\delta t) - \frac{1}{2}\mathbf{a}(t+\delta t)\delta t + \frac{1}{4}\mathbf{v}_i''(t+\delta t)\delta t^2 \\ &\approx \mathbf{v}_i(t+\delta t) - \frac{1}{2}\frac{\mathbf{F}_i(t+\delta t)}{m_i}\delta t + \frac{1}{4}\mathbf{v}_i''(t+\delta t)\delta t^2 \\ \mathbf{v}_i(t+\delta t) &\approx \mathbf{v}_i(t+\frac{1}{2}\delta t) + \frac{1}{2}\frac{\mathbf{F}_i(t+\delta t)}{m_i}\delta t - \frac{1}{4}\mathbf{v}_i''(t+\delta t)\delta t^2\end{aligned}$$

$$\boxed{\mathbf{v}_i(t+\delta t) = \mathbf{v}_i(t) + \frac{1}{2m_i}\delta t\left(\mathbf{F}_i(t) + \mathbf{F}_i(t+\delta t)\right) + O(\delta t^3)}$$

Newton 운동방정식의 적분
-Velocity Verlet algorithm summary

- Advancing the position of a particle

$$\mathbf{r}_i(t+\delta t) = \mathbf{r}_i(t) + \mathbf{v}_i(t)\delta t + \frac{1}{2}\frac{\mathbf{F}_i(t)}{m_i}\delta t^2 + O(\delta t^3)$$

- Advancing the velocity of a particle

$$\mathbf{v}_i(t+\delta t) = \mathbf{v}_i(t) + \frac{1}{2m_i}\delta t\left(\mathbf{F}_i(t) + \mathbf{F}_i(t+\delta t)\right) + O(\delta t^3)$$

Application of velocity Verlet algorithm to a harmonic oscillator model

- Newtonian equation of motion for a harmonic oscillator

$$m_a\frac{d^2x_a}{dt^2} = -kx_a \Rightarrow x_a(t) = A\cos(\omega t) + B\sin(\omega t) \qquad (\omega = \sqrt{k/m_a})$$

만약 $x_a(0) = 0$, $v_a(0) = v_0$ 라면,

$$x_a(t) = \frac{v_0}{\omega}\sin(\omega t)$$

- 예: $k = 4,\ m_a = 1,\ v_0 = 1 \Rightarrow \omega = 2$

해석해: $x_a(t) = \frac{1}{2}\sin(2t), \qquad v_a(t) = \cos(2t)$

수치해:
$$\begin{aligned} x_a(t+\delta t) &= x_a(t) + v_a(t)\delta t + \frac{1}{2}\frac{F_a(t)}{m}\delta t^2 \\ &= x_a(t) + v_a(t)\delta t - 2x_a(t)\delta t^2 \\ v_i(t+\delta t) &= v_i(t) + \frac{1}{2m_i}\left(F_i(t) + F_i(t+\delta t)\right)\delta t \\ &= v_i(t) - \left(2x_a(t) + 2x_a(t+\delta t)\right)\delta t \end{aligned}$$

앙상블 (Ensemble)

- 앙상블: 미세상태의 집합
- 앙상블의 종류
  1) 소정준 앙상블 (microcanonical ensemble): 고립계에 대응하는 앙상블. 입자의 개수 (N), 계의 부피 (V), 계의 에너지 (E)는 일정.
  2) 정준 앙상블 (canonical ensemble): 닫힌계에 대응하는 앙상블. 입자의 개수 (N), 계의 부피 (V), 계의 온도 (T)가 일정.
  3) 대정준 앙상블 (grand canonical ensemble): 열린계에 대응하는 앙상블. 화학퍼텐셜 (μ), 계의 부피 (V), 계의 온도 (T)가 일정
  4) 등온-등압 앙상블: 부피가 일정하지 않은 닫힌계에 대응하는 앙상블. 입자의 개수 (N), 계의 압력 (p), 계의 온도 (T)는 일정.
- 앙상블 평균 (ensemble average)는 시간 평균 (time average)과 동등하다.
- 분자동역학 모의실험은 계의 열역학적 성질에 대해 시간 평균을 계산한다.
- Monte Carlo 모의실험은 같은 성질에 대해 앙상블 평균을 계산한다.

초기조건에 대한 Maxwell-Boltzmann 분포

- X 방향 속도성분에 대한 Maxwell-Boltzmann 분포

$$f(v_x)dv_x = \left(\frac{m}{2\pi k_B T}\right)^{1/2} \exp\left[-\frac{mv_x^2}{2k_B T}\right]dv_x$$

- 3차원 속도에 대한 Maxwell-Boltzmann 분포

$$f(v_x)f(v_x)f(v_x)dv_x dv_y dv_z = \left(\frac{m}{2\pi k_B T}\right)^{3/2} \exp\left[-\frac{m(v_x^2+v_y^2+v_z^2)}{2k_B T}\right]dv_x dv_y dv_z$$

$$= \left(\frac{m}{2\pi k_B T}\right)^{3/2} \exp\left[-\frac{mv^2}{2k_B T}\right]dv_x dv_y dv_z$$

기체 운동에 대한 동역학 모델

- Δt 동안의 총 운동량 변화

$$\text{momentum change} = \text{impact} = \frac{NAv_x\Delta t}{2V} \times 2mv_x$$
$$= \frac{NAmv_x^2\Delta t}{V}$$

- 압력

$$P = \frac{\text{impact}}{A\Delta t} = \frac{Nm\langle v_x^2\rangle}{V}$$

- x 방향의 분자 당 평균 운동에너지

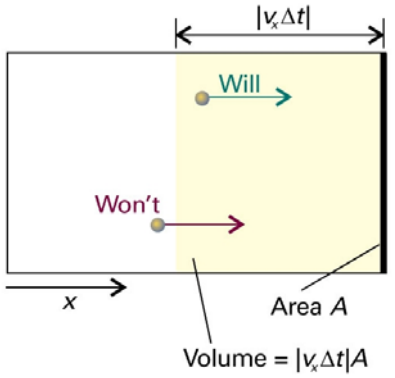


Fig. 21.2 A molecule will reach the wall on the right within an interval Δt if it is within a distance $v_x\Delta t$ of the wall and travelling to the right.

$$\text{x 방향의 분자 당 평균 운동에너지} = \frac{1}{2}m\langle v_x^2\rangle = \frac{1}{2}\frac{PV}{N} = \frac{1}{2}\frac{Nk_BT}{N} = \frac{1}{2}k_BT$$

Velocity rescaling

- Average kinetic energy of a molecule

$$\text{average KE of a molecule} = \frac{1}{2}m\langle v^2\rangle = \frac{1}{2}m\langle v_x^2 + v_y^2 + v_z^2\rangle$$
$$= \frac{1}{2}m\langle v_x^2\rangle + \frac{1}{2}m\langle v_y^2\rangle + \frac{1}{2}m\langle v_z^2\rangle = \frac{1}{2}k_BT + \frac{1}{2}k_BT + \frac{1}{2}k_BT = \frac{3}{2}k_BT$$

- Scaling factor

$$\frac{1}{2}m\sum_{n=1}^{N}\langle (v_n^i)^2\rangle = \frac{3}{2}(N-1)k_BT$$
$$\frac{1}{2}m\sum_{i=1}^{N}\langle (v_n^f)^2\rangle^f = \frac{3}{2}(N-1)k_BT_{ref}$$

$$\sum_{n=1}^{N}\langle (v_n^f)^2\rangle = \frac{(N-1)T_{ref}}{(N-1)T}\sum_{n=1}^{N}\langle (v_n^i)^2\rangle$$
$$= \sum_{n=1}^{N}\left\langle \left\{ \left(\frac{(N-1)T_{ref}}{(N-1)T}\right)^{1/2} v_n^i \right\}^2 \right\rangle$$
$$= \sum_{i=1}^{N}\left\langle \left\{ \left(\frac{3(N-1)T_{ref}}{m\sum_{n=1}^{N}\langle (v_n^i)^2\rangle}\right)^{1/2} v_n^i \right\}^2 \right\rangle$$

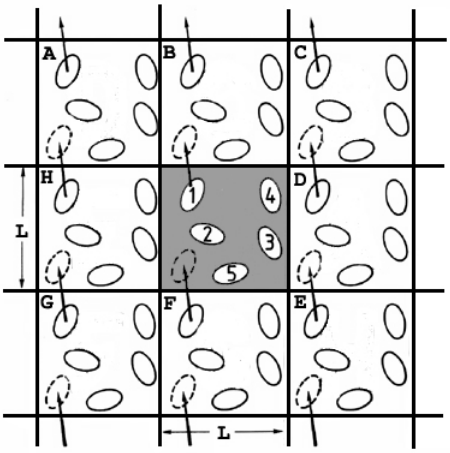
$$\text{scaling factor} = \left(\frac{3(N-1)T_{ref}}{m\sum_{n=1}^{N}\langle (v_n^i)^2\rangle}\right)^{1/2}$$

Temperature control methods

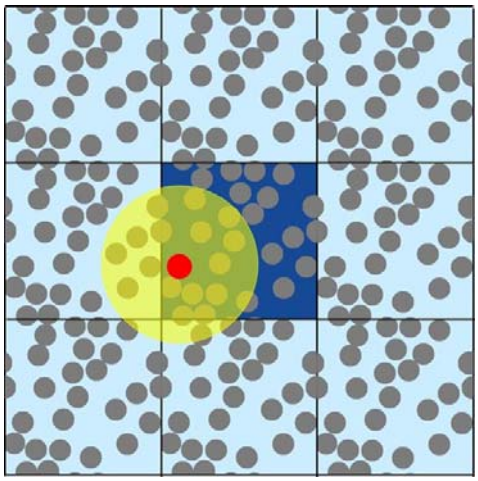
1. Velocity rescaling method
2. Nose-Hoover method
3. Temperature reassignment methods
4. Lowe-Anderson thermostat method
5. **Langevin thermostat method**

Periodic boundary conditions and minimum image convention

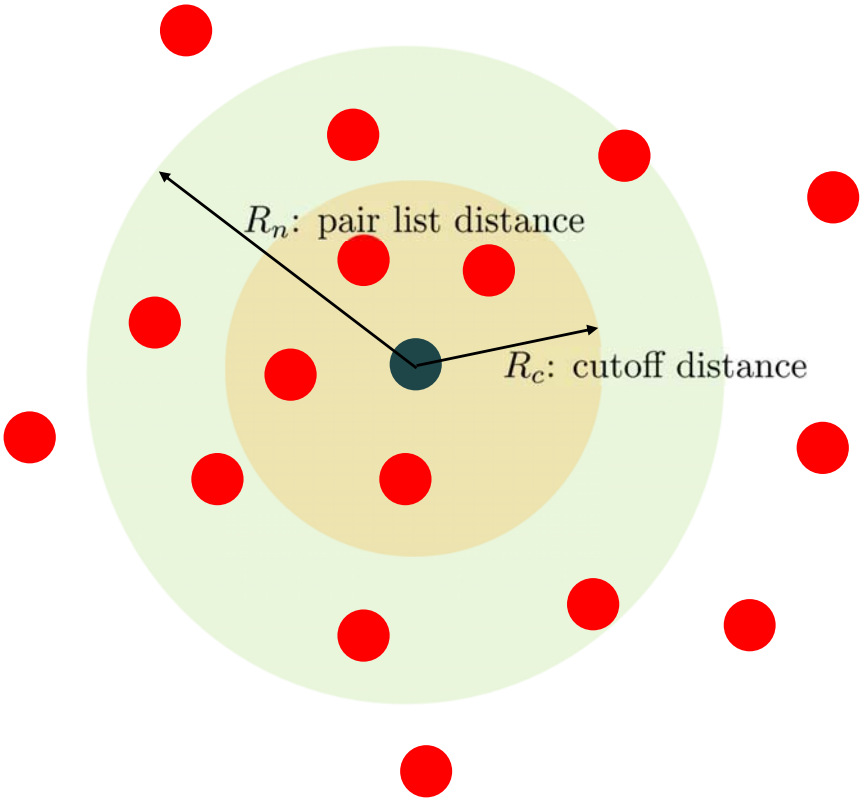
- Periodic boundary conditions (PBC)



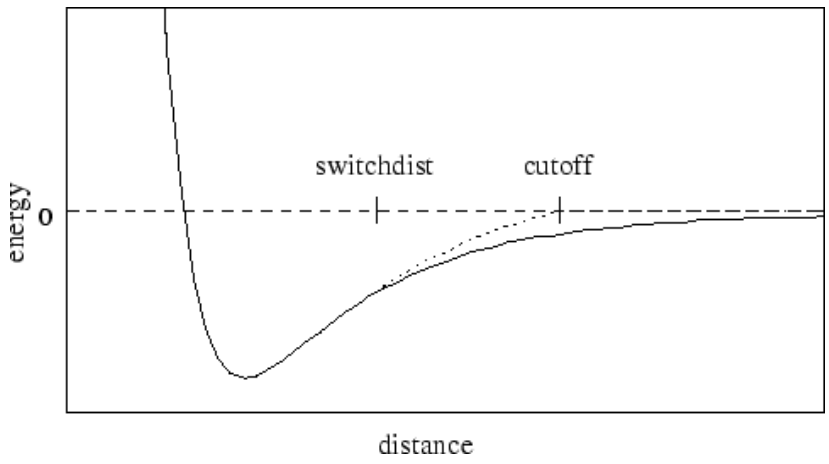
- Minimum Image Convention (MIC)



Cutoff distance and neighbor list



Switching function



$$V^{LJ}(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N) = \sum_{i=1}^{N} \sum_{j>i}^{N} 4\epsilon \left[\left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^{6} \right] f(r_{ij})$$

where

$$f(r) = \begin{cases} 1 & r \le r_{sd} \\ \frac{(r_c^2 - r^2)^2 \left[3(r_c^2 - r_{sd}^2) - 2(r_c^2 - r^2) \right]}{(r_c^2 - r_{sd}^2)^3} & r_{sd} < r \le r_c \\ 0 & r > r_c \end{cases}$$